

Hallucinations & the Fact-Checking Checklist

What Is an AI Hallucination?

An AI hallucination is a confident, fluent, but factually false output from an LLM. The model generates text by predicting statistically likely next tokens — not by verifying facts against a trusted database. When it lacks reliable training signal, it fabricates plausible-sounding content.

Critical insight: hallucinations are a feature of the architecture, not a bug that can simply be 'fixed' — they require systematic mitigation.

Why Hallucinations Are Dangerous

- Legal: lawyers have submitted AI-generated citations to courts that turned out to be fabricated cases.
- Medical: patients following AI medical advice without verification risk serious harm.
- Academic: hallucinated statistics and sources erode the integrity of research.
- Business: incorrect market data or contract terms generated by AI can cause financial loss.

The Three-Step Fact-Checking Checklist

Apply this checklist before acting on any AI-generated factual claim:

Step	Action	Example
1. IDENTIFY	Flag every specific factual claim: names, dates, statistics, citations	'The study from Harvard (2021) found...' — flag this.
2. SEARCH	Independently verify each flagged claim using authoritative sources	Search Pubmed, Google Scholar, official government sites.
3. CROSS-REFERENCE	Confirm the claim appears in at least two independent, credible sources	— or none — treat as unverified.

Reducing Hallucinations — Developer Strategies

- RAG (Retrieval-Augmented Generation): ground responses in retrieved, verified documents.
- Temperature tuning: lower temperature = less creative, more conservative and factual outputs.
- Structured prompts: 'Answer only based on the context below. If unsure, say so.'
- Citation requirements: prompt the model to cite sources for every claim.
- Uncertainty prompting: ask the model to rate its own confidence.

Key Takeaways

- Hallucinations are confident false outputs — a structural feature of LLMs.
- High-stakes domains (law, medicine, finance) require rigorous fact-checking.
- The three-step checklist: Identify → Search → Cross-reference.
- Developer mitigations: RAG, temperature control, structured prompts, citation requirements.

Quick-Revision Questions

- Why do LLMs hallucinate? (architectural explanation)
- Apply the three-step checklist to an AI claim about a historical statistic.
- Name three developer-side strategies to reduce hallucination.